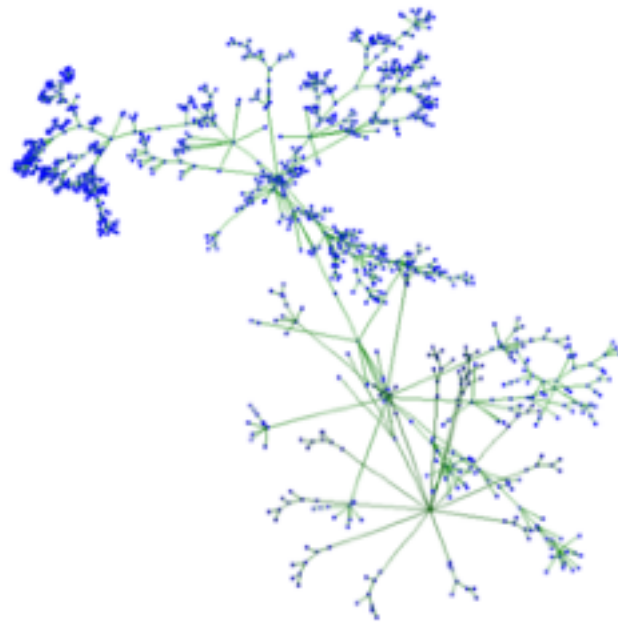


100% accurate, style-preserving transformation because of a tech leap in code representation

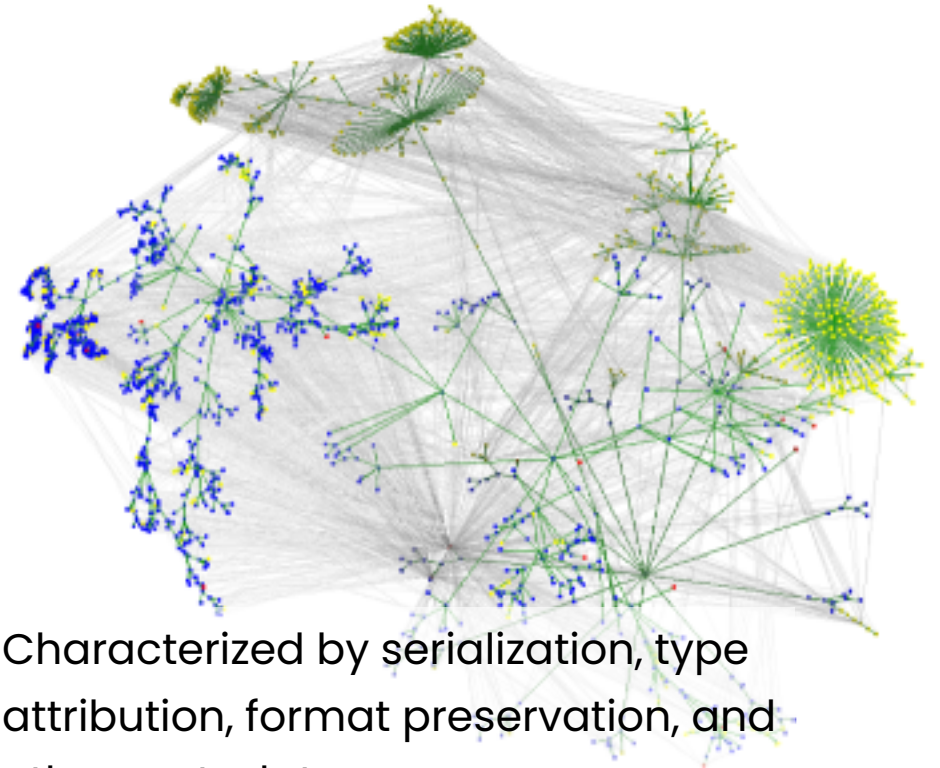
Text-based search

```
void m() {  
    if(c > 2) {  
        x = c;  
    }  
    while(c < 10) {  
        x += p();  
        c++;  
    }  
}
```

Abstract Syntax Tree (AST)



Lossless Semantic Tree (LST)



Characterized by serialization, type attribution, format preservation, and other metadata

+ another 600,000 nodes that didn't fit in this graphic

Imperative Recipe API

```
public class MakePublic extends Recipe {  
    public String getDisplayName() {  
        return "Make Class Public";  
    }  
    @Override  
    protected JavaVisitor<ExecutionContext> getVisitor() {  
        return new ChangeTypeVisitor();  
    }  
    private class MakePublicVisitor  
        extends JavaVisitor<ExecutionContext> { /* .. */ }  
}  
class JavaVisitor<P> extends TreeVisitor<J, P> {  
    J visitStatement(Statement statement) {}  
    J visitAnnotatedType(J.AnnotatedType annotatedType) {}  
    J visitAnnotation(J.Annotation annotation) {}  
    J visitAssert(J.Assert azzert) {}  
    J visitAssignment(J.Assignment assign) {}  
    J visitAssignmentOperation(J.AssignmentOperation assignOp) {}  
    // ...  
}
```

